

RT 508

GEOSPATIAL TECHNOLOGIES IN AGRO- RURAL SECTOR

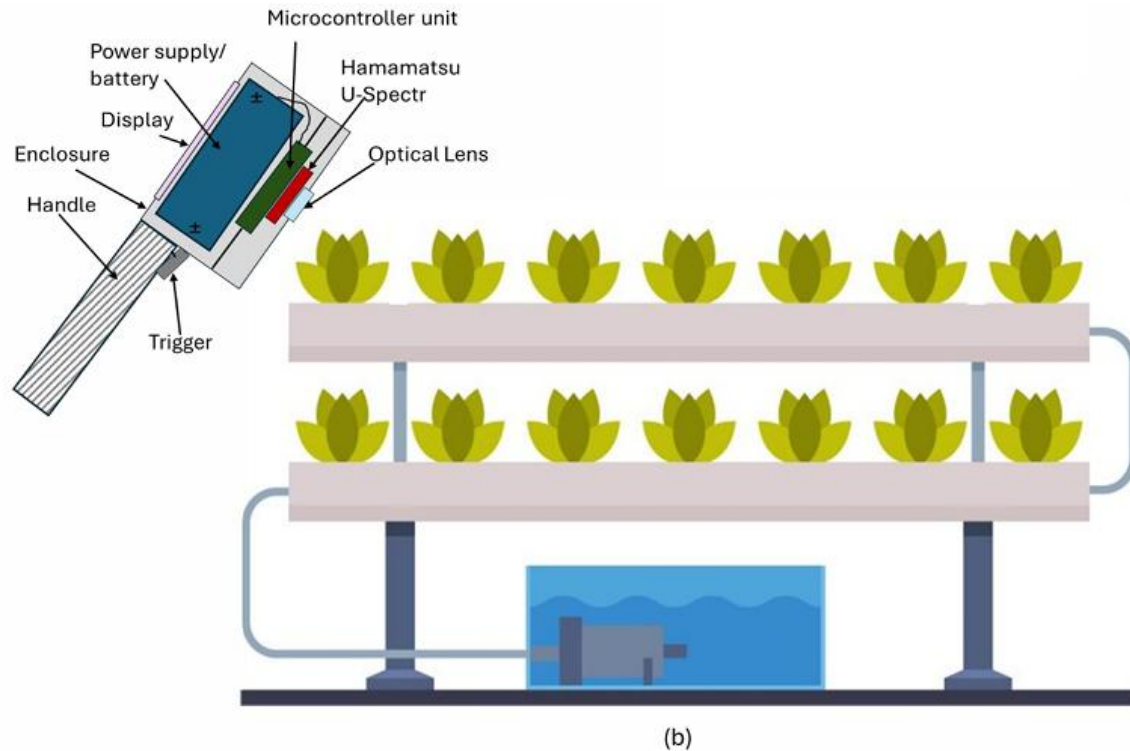
**MINI PROJECT: Developing a handheld spectral device for
leaf chlorophyll content measurement**

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INTRODUCTION



CropHySense: An Affordable Hyperspectral Sensing Device for Precision Plant Stress Monitoring in Agricultural Crops

Spectral device: Analytical instrument to measure interaction of light with a material across different wavelengths to determine its physical or chemical composition

OBJECTIVES

- To design and develop a portable hyperspectral sensing device
- To measure leaf reflectance spectra and estimate chlorophyll content
- To ensure low cost, user-friendliness, and accuracy comparable to commercial devices

Major Steps:

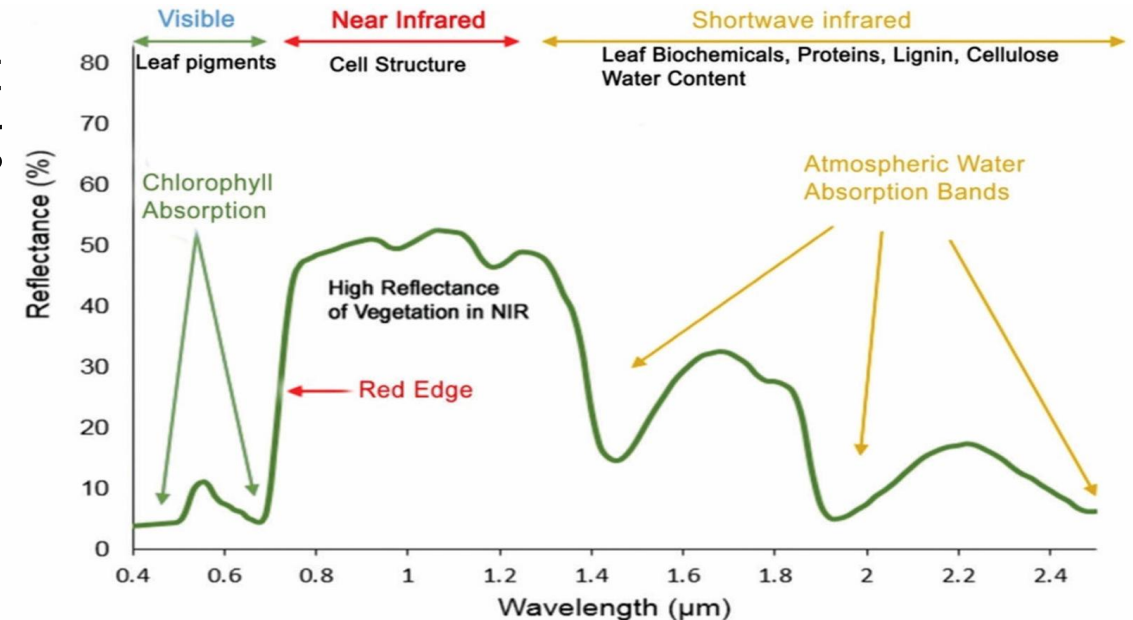
- Understand the working of the sensor and microcontroller
- Operate Arduino and Processing software for data acquisition
- Write code to obtain and analyze digital reflectance data
- Plot and interpret spectral graphs for chlorophyll estimation
- Design and assemble a compact, enclosed handheld device

THEORY

Chlorophyll (Chl): Green pigment responsible for photosynthesis, converting light energy into chemical energy.

Chl content provides critical indicators of:

- Plant Vigor and Health Status
- Nutrient Availability
- Biological Stress Levels

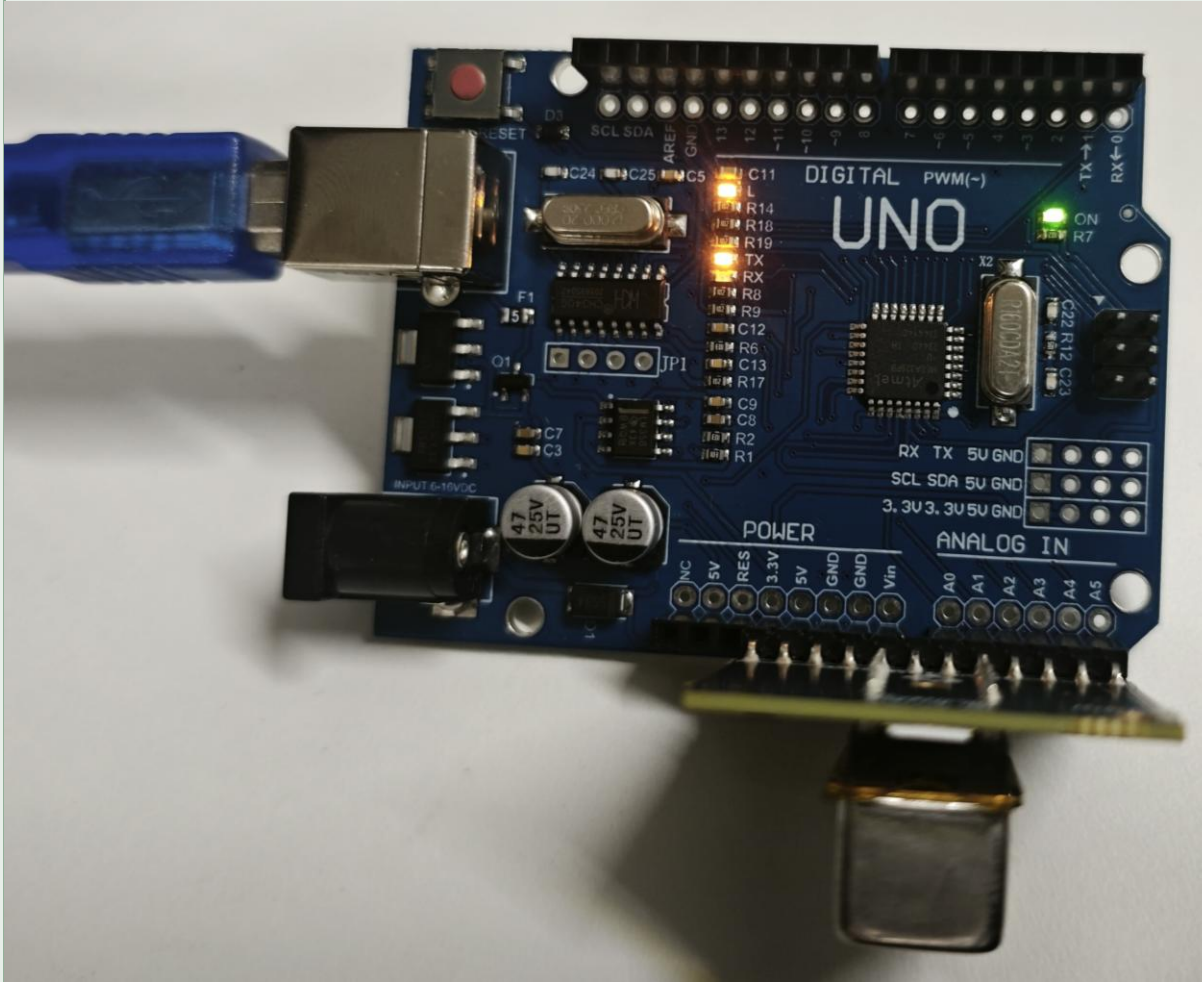


The spectral reflectance curve of vegetation

(Roman et al., 2018) Fig. 90. The spectral reflectance curve of vegetation.

Data obtained enables precise optimization of fertilization schedules,
facilitates the early identification of crop distress
Contributes to maximizing agricultural yield and product quality

SYSTEM DESIGN



- Hamamatsu C12880MA (Sensor)
- Arduino UNO (Microcontroller Board)
- USB Type-A to Type-B Cable
- Arduino Software (Arduino IDE)
- Processing Software
- Microsoft Excel

Hamamatsu C12880MA (Sensor)

Ultra-compact, high-sensitivity MEMS
micro-spectrometer for integration into
portable and embedded systems

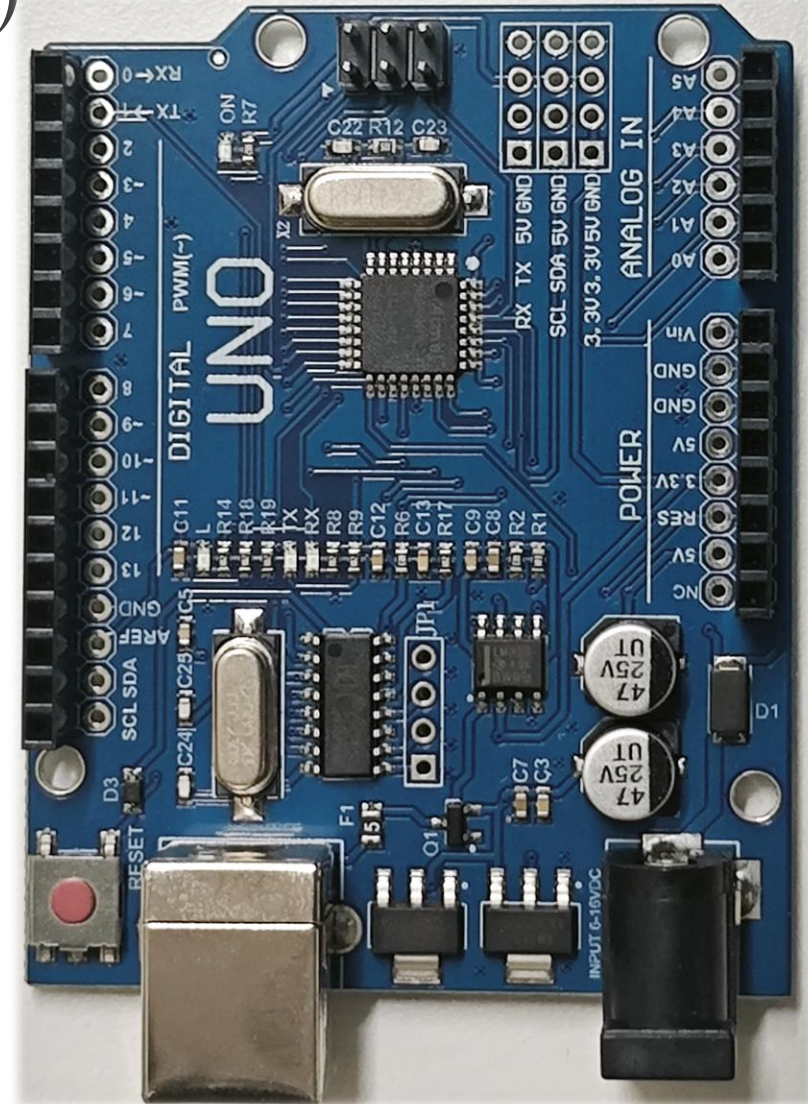
- Spectral response range: **350–850 nm**
- Spectral resolution: **15 nm**
- Number of pixels: **288**
- Dimensions: **20.1 × 12.5 × 10.1 mm**
- Weight: **5 g** (finger-tip sized)
- Interface: **SPI communication**
- Trigger compatible: supports
synchronized integration
- Packaging: hermetically sealed for high
humidity resistance



Arduino UNO (Microcontroller Board)

Microcontroller-based open-source development board used for building and controlling electronic systems

- Microcontroller: ATmega328P
- Operating voltage: 5 V
- Digital I/O pins: 14 (6 support PWM output)
- Analog input pins: 6
- Clock speed: 16 MHz
- Communication interfaces: USB, UART (TX/RX), SPI, and I2C
- Programming platform: Arduino IDE — uses C/C++-based language



USB Type-A to Type-B Cable

Connects Arduino UNO to a computer for power supply and data transfer

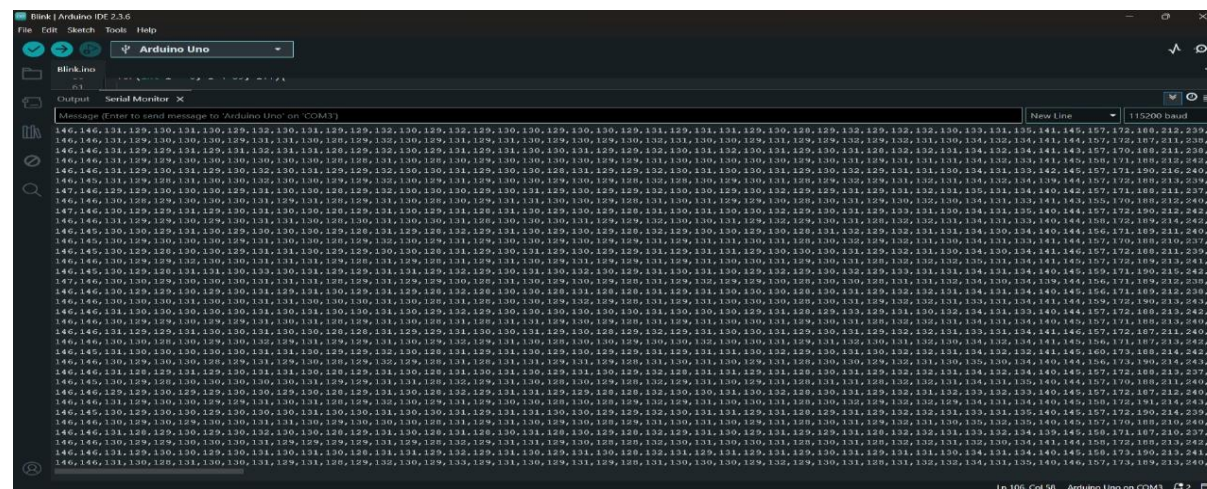
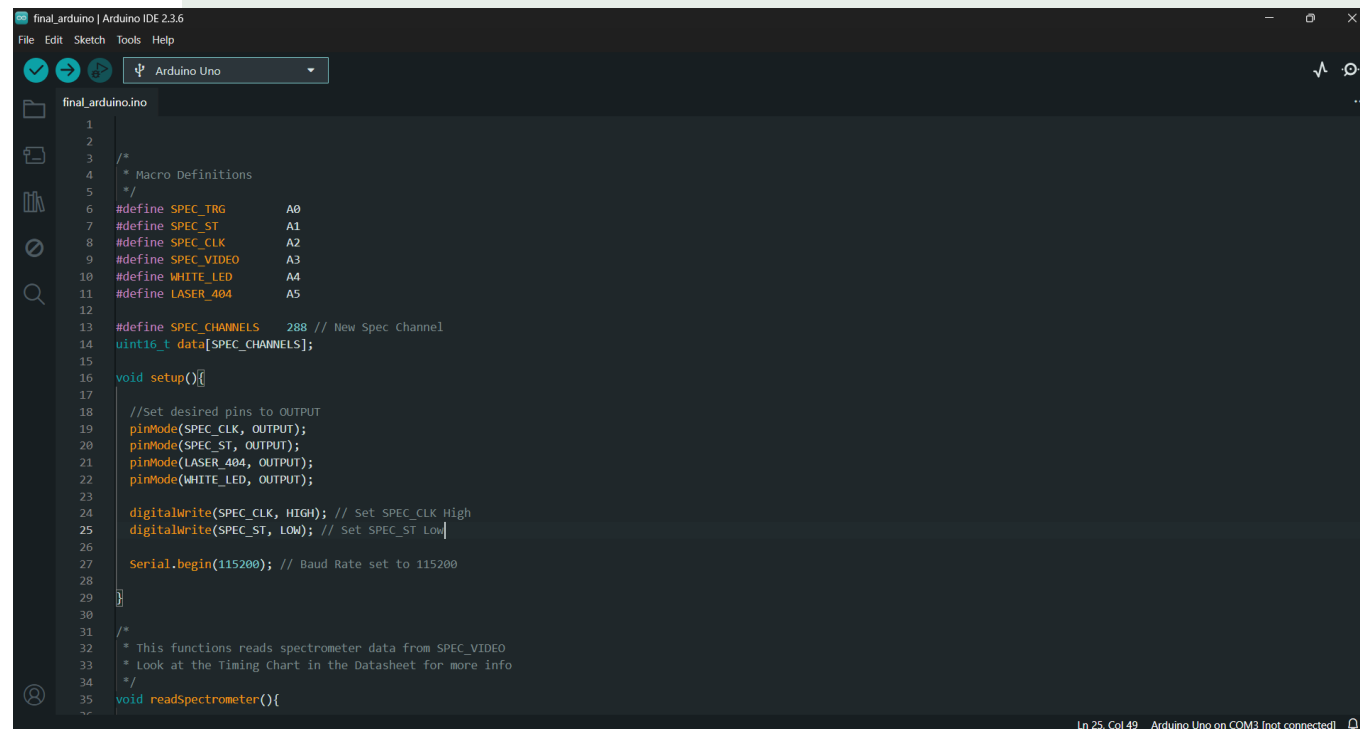
Material:

- Outer layer: **Flexible PVC** insulation (durability and protection)
- Internal conductors: **Copper wires**
- Shielding: **Braided aluminum foil** or **metal mesh** to reduce electromagnetic interference



Arduino UNO

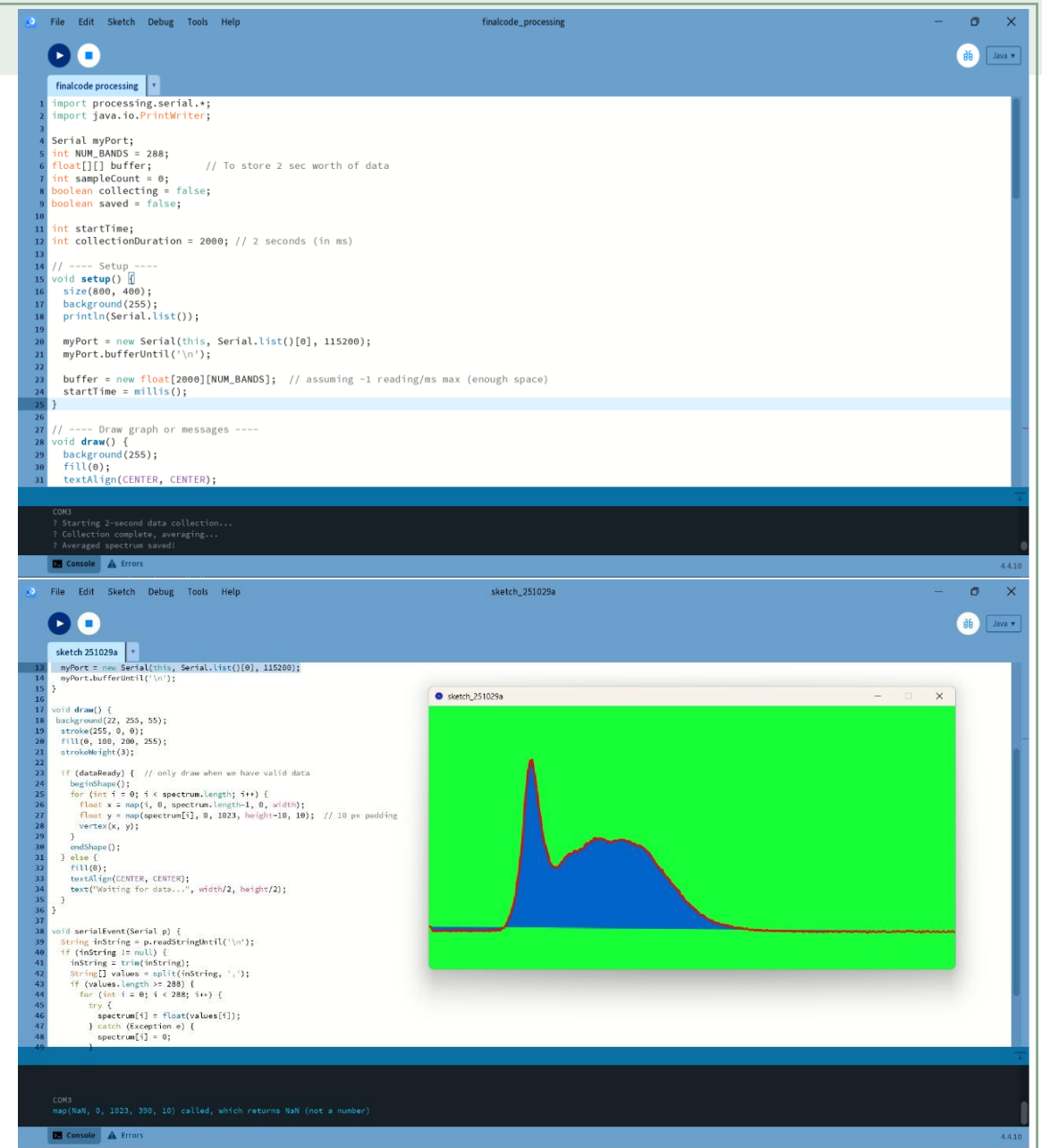
- Microcontroller Board (Hardware)
- Role: Controls and processes signals from the sensor
- Function: Interfaces with spectral sensor, executes code, and sends data to software (e.g., Processing or PC)
- Category: Hardware / Control Unit



Processing Software

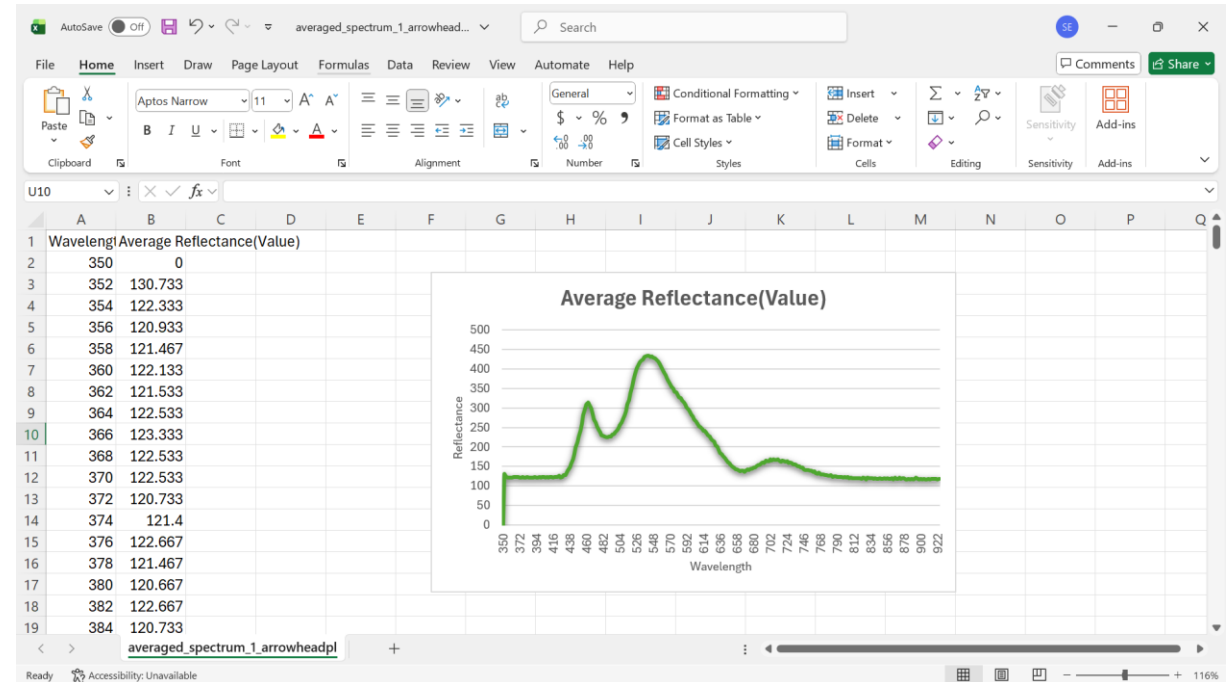
Visualization and Data Processing Software

- Used to visualize sensor data graphically
- Provides interactive interface to analyse data received from Arduino via serial communication
- Used for graph plotting and pattern recognition in spectral data



Excel Sheet (Microsoft Excel)

- Used to store, organize, and analyze numerical data collected from experiments
- Enables plotting graphs, calculating averages, and performing statistical analysis of reflectance data
- Supports data export from Arduino or Processing for further analysis

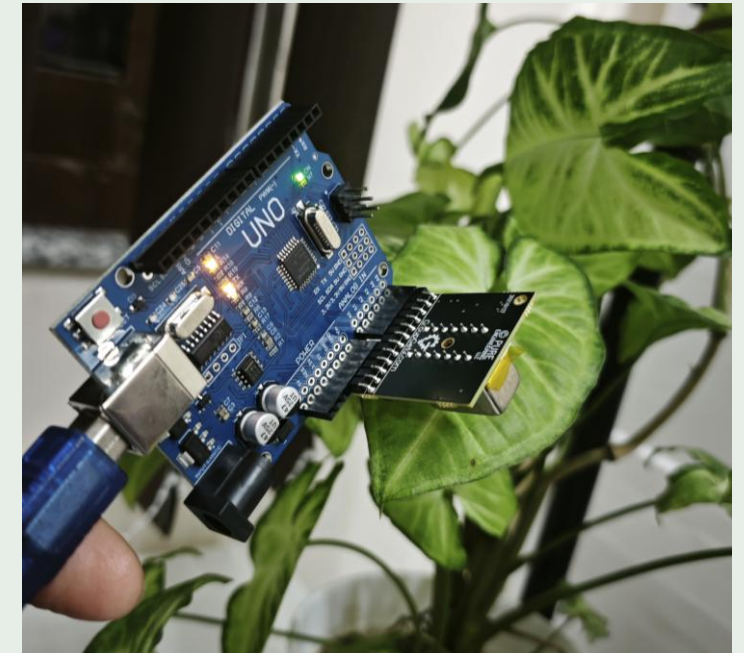
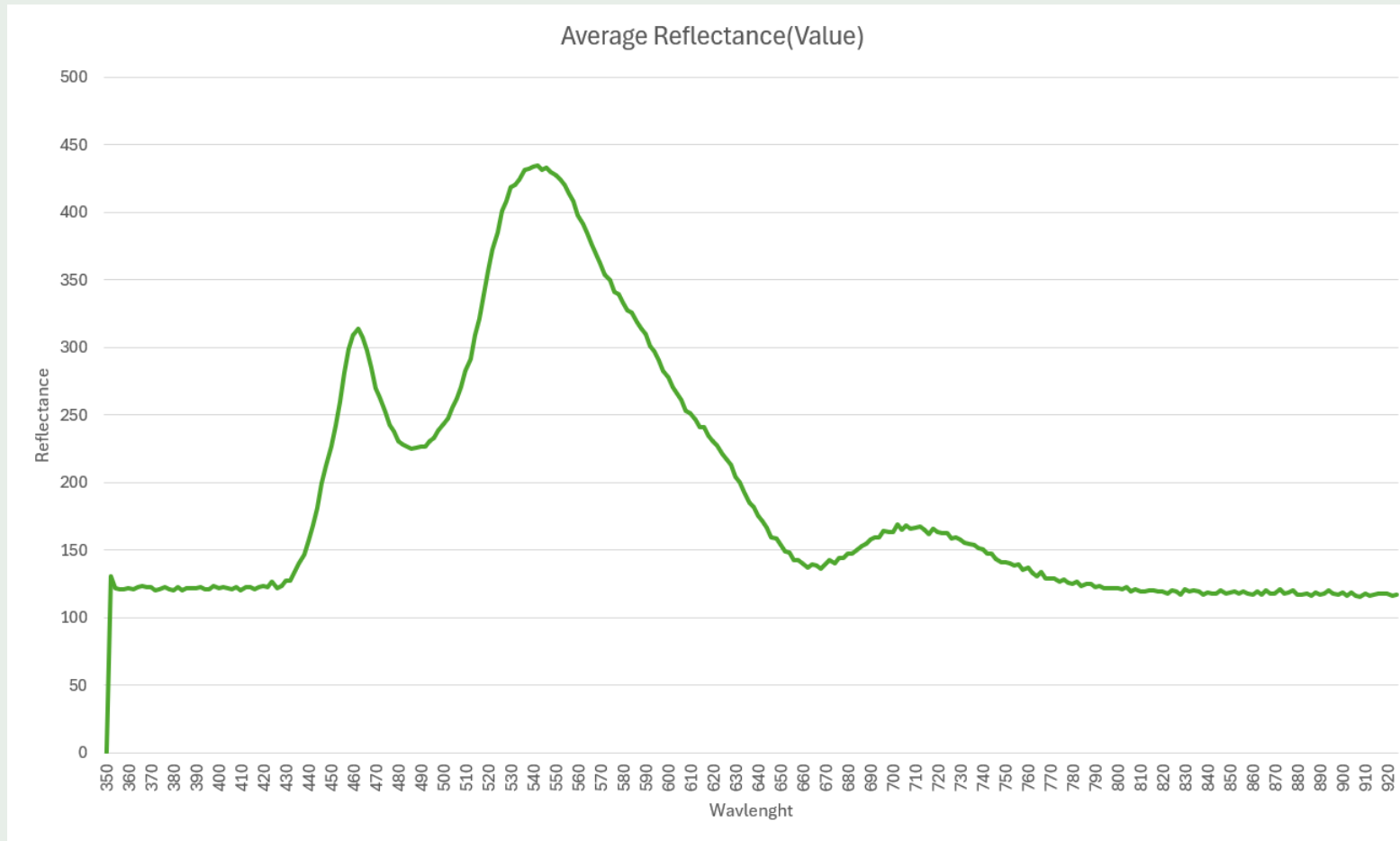


OBSERVATIONS

- Reflectance data captured across UV-visible-NIR transition regions
- All plant species exhibited characteristic **vegetation spectral signature**
- Succulent plants (Jade) showed higher visible reflectance than thin-leaved species

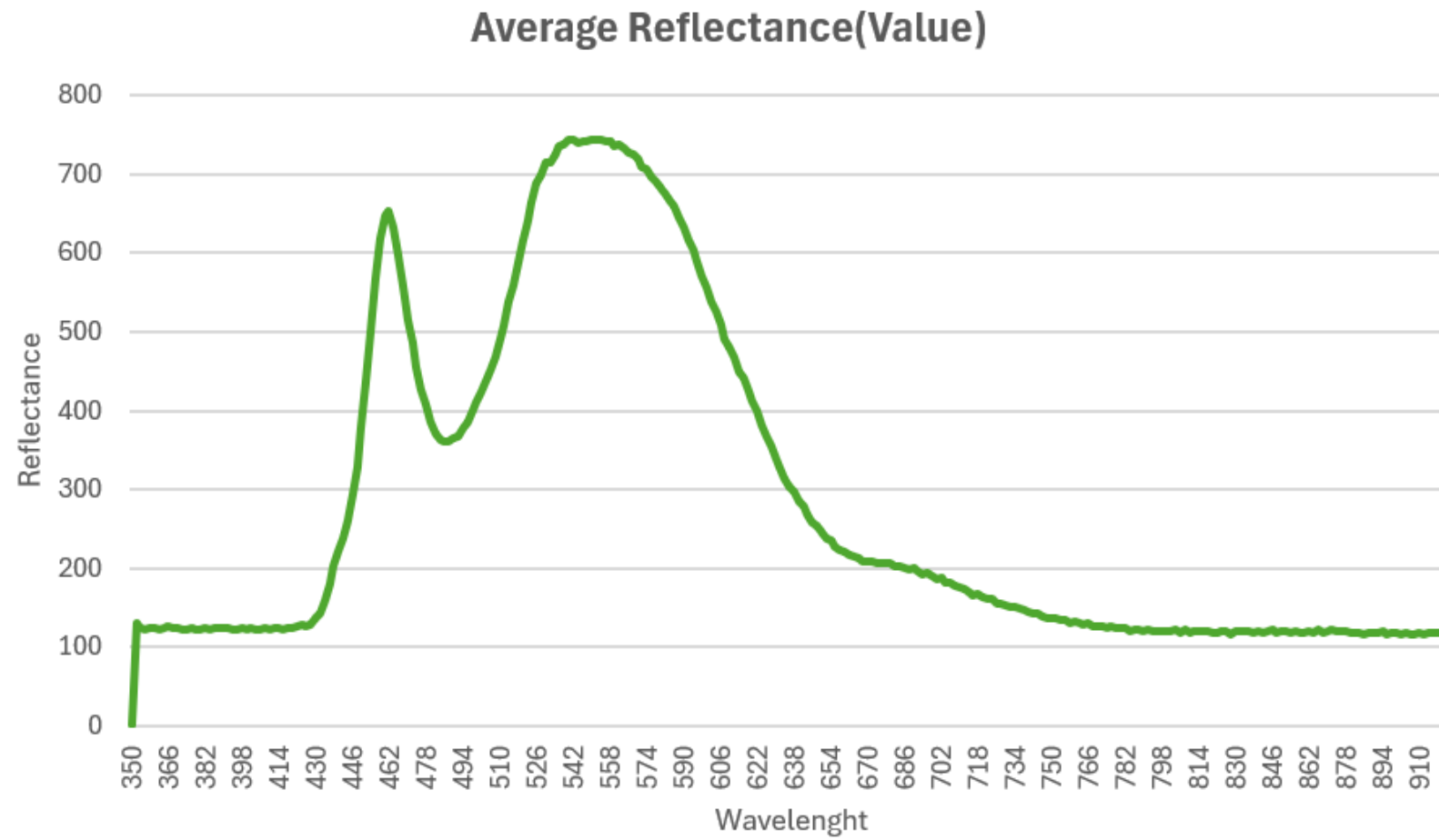


ARROWHEAD



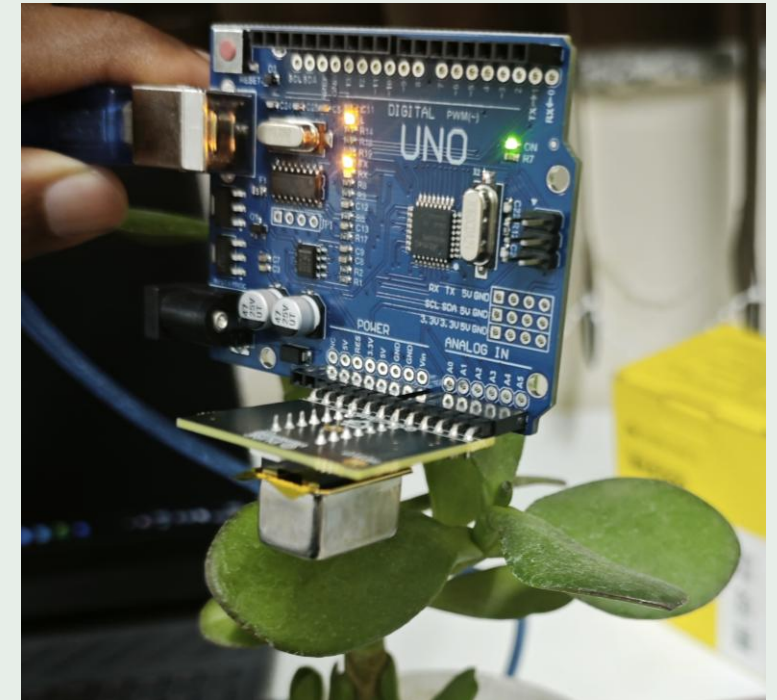
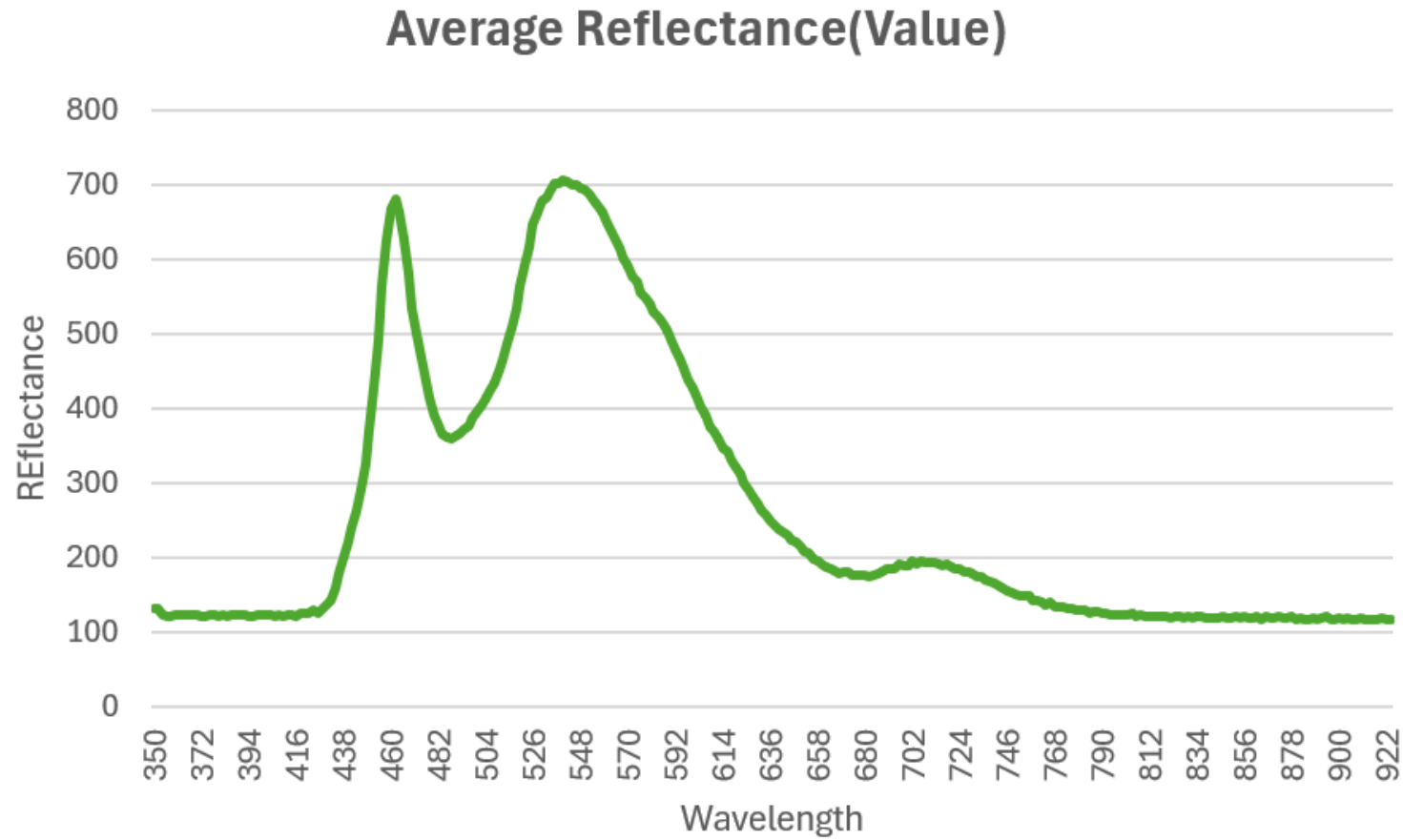
Wavelength(nm)	Average Reflectance(Value)
350	0
352	130.73334
354	122.333336
356	120.933334
358	121.46667
360	122.13333
362	121.53333

CHINESE EVERGREEN



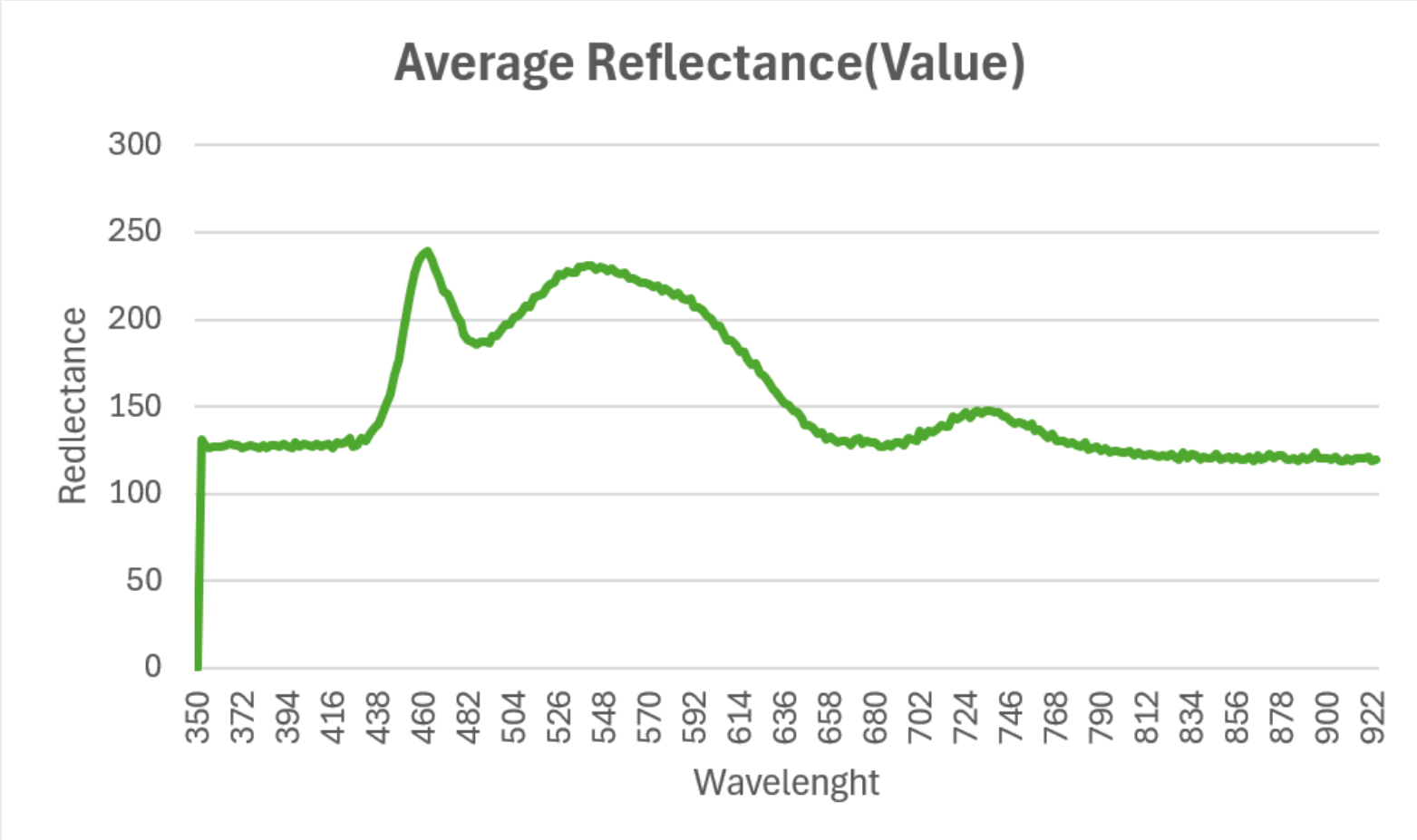
Wavelength(nm)	Average Reflectance(Value)
350	0
352	131
354	124.2
356	122.46667
358	123.666664
360	124.26667
362	123.26667

JADE PLANT



Wavelength(nm)	Average Reflectance(Value)
350	131
352	130.93333
354	122.6
356	121.333336
358	121.4
360	122
362	121.73333

ZEBRA PLANT

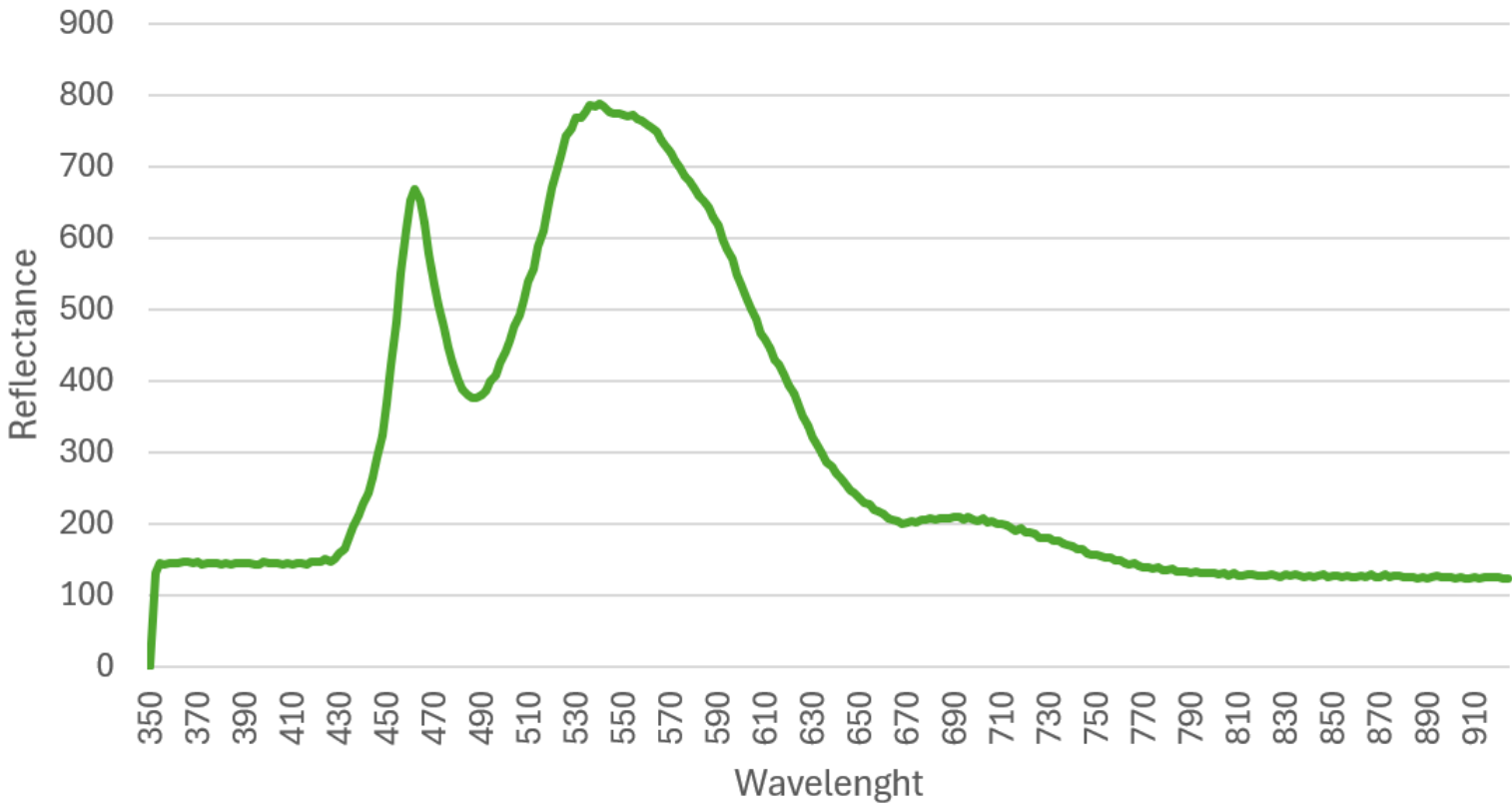


Wavelength(nm)	Average Reflectance(Value)
350	0
352	130.8
354	127.13333
356	125.86667
358	126.933334
360	127.333336
362	126.933334

WHITE ARALI



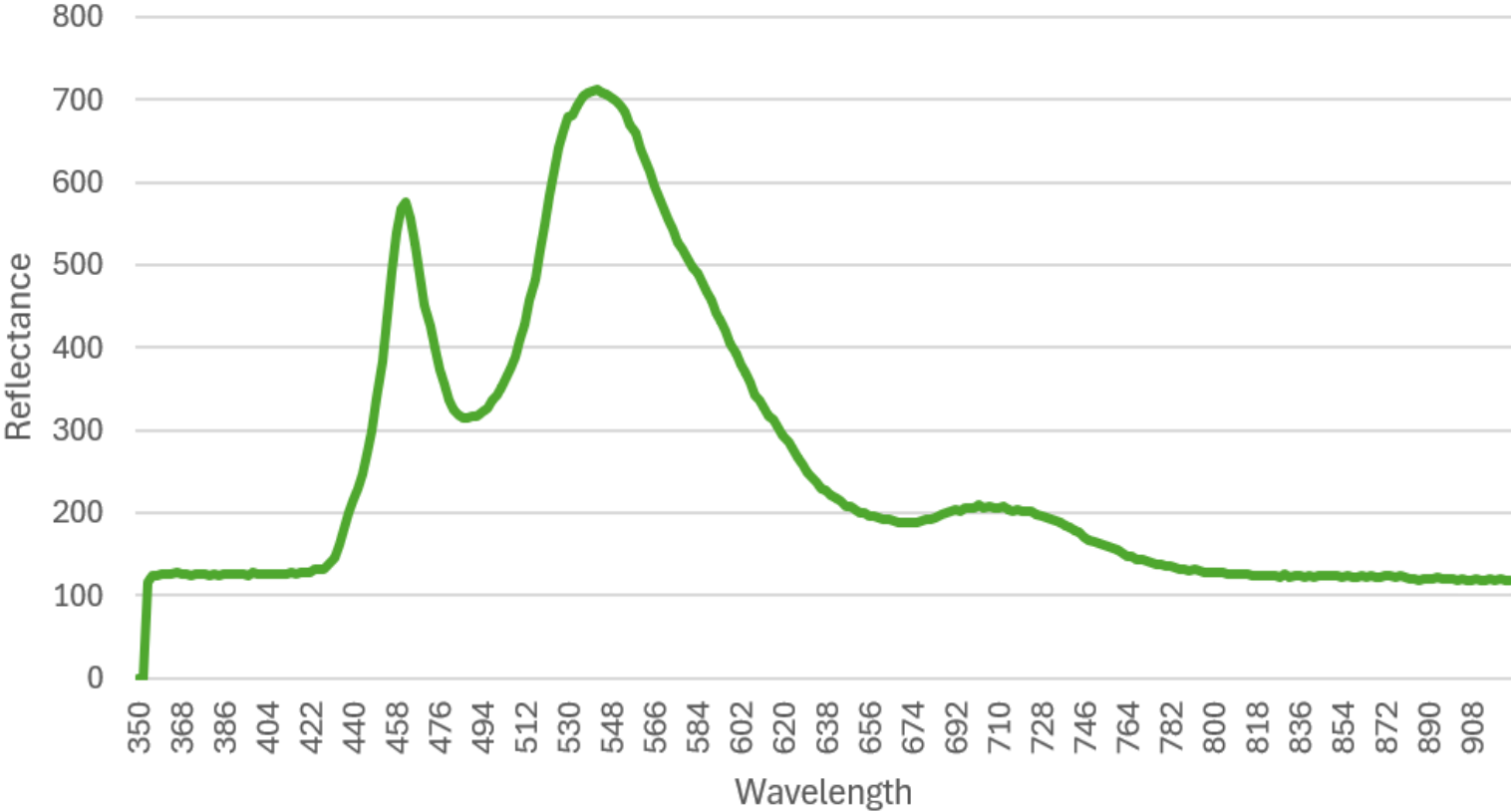
Average Reflectance(Value)



Wavelength(nm)	Average Reflectance(Value)
350	0
352	130.93333
354	145.6
356	144.13333
358	145.46666
360	145.8
362	145

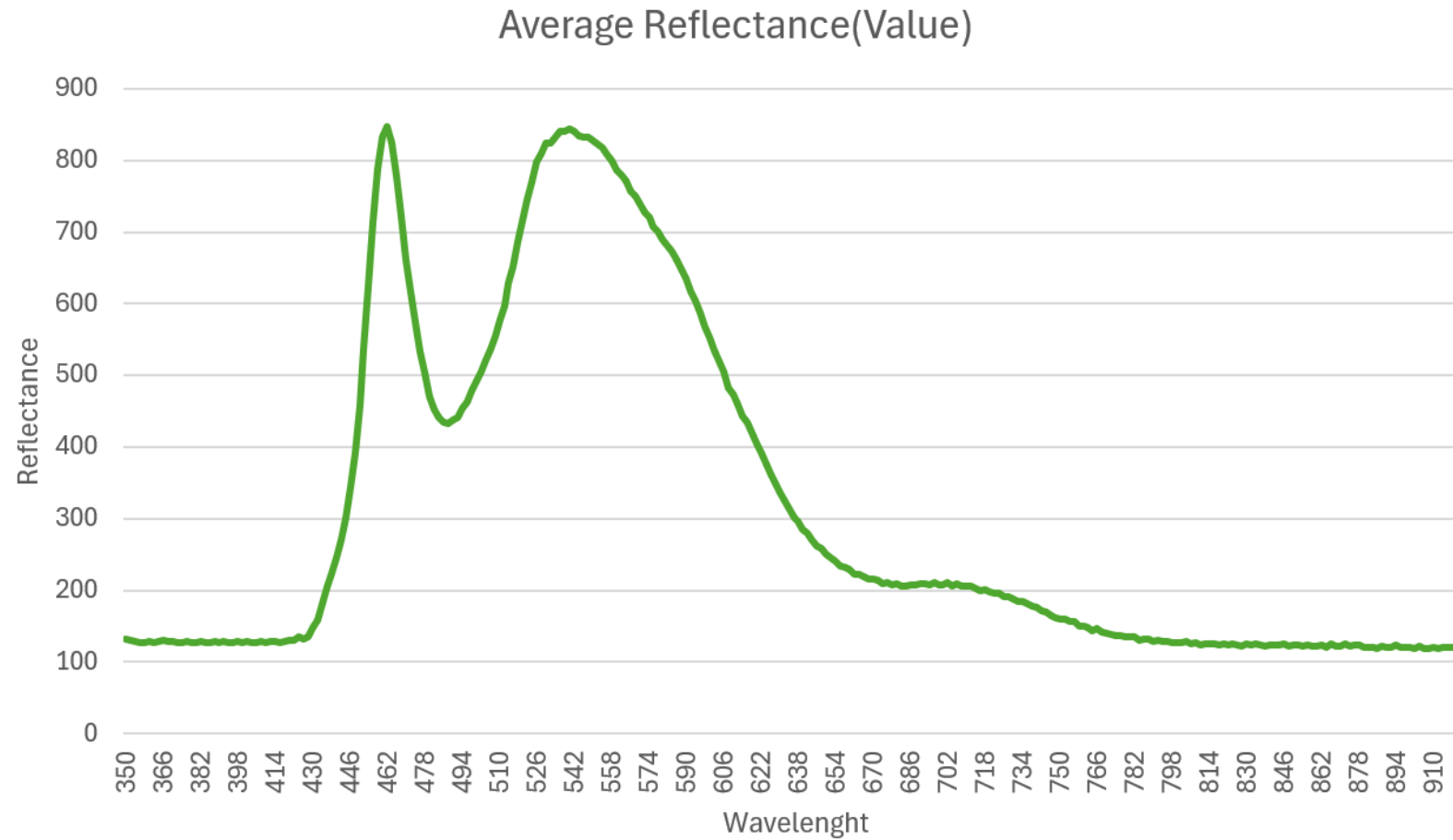
MONEY PLANT

Average Reflectance(Value)



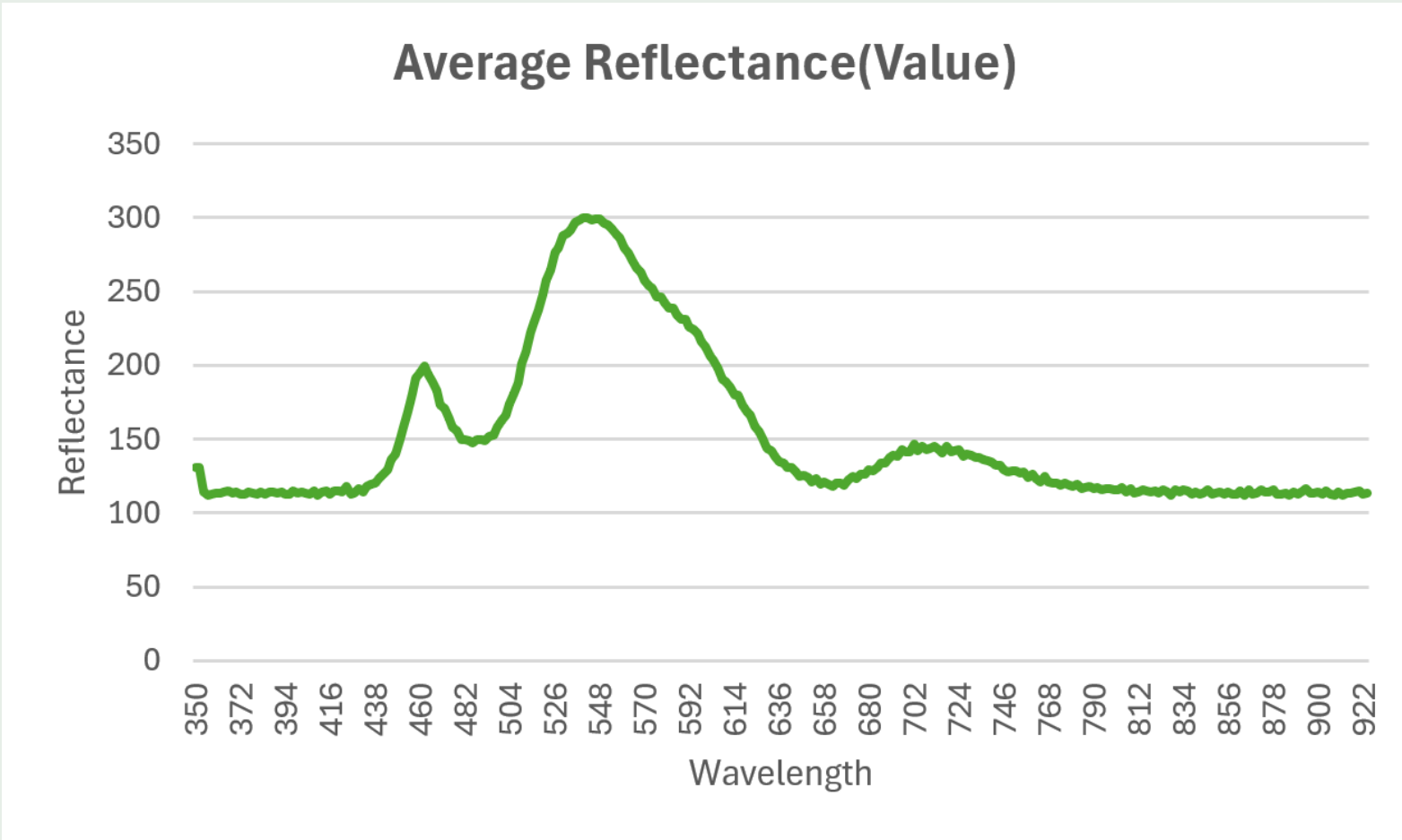
Wavelength(nm)	Average Reflectance(Value)
350	0
352	0
354	116.933334
356	124.86667
358	124.46667
360	126.46667
362	125.53333

PARLOUR PALM



Wavelength(nm)	Average Reflectance(Value)
350	131.4
352	130.93333
354	128.93333
356	127.06666
358	127.53333
360	128.46666
362	127.13333

BABY RUBBER PLANT



Wavelength(nm)	Average Reflectance(Value)
350	131.26666
352	130.86667
354	114.333336
356	112.13333
358	113.13333
360	113.86667
362	113.2
364	114.333336



FUTURE ASPECTS

- 3D-printed model to enclose the sensor and microcontroller unit
- Development of a low-cost, portable handheld spectrophotometer
- Application of the PROSAIL model to estimate chlorophyll content
- Integration with mobile app or Bluetooth for real-time data transfer
- Extension of wavelength range to include UV and NIR regions
- Optimization of materials and miniaturization for better durability and portability

CONCLUSION

- Successfully designed low-cost spectrophotometer for chlorophyll estimation
- Hamamatsu C12880MA sensor integrated with Arduino UNO provided accurate spectral measurements across 340–850 nm range
- Device demonstrated capability to capture characteristic vegetation reflectance patterns with clear chlorophyll absorption features
- Spectral data successfully acquired and visualized using Arduino IDE and Processing software for eight different plant species
- The system can evolve into a reliable, compact, and farmer-friendly spectrometer for real-time plant health monitoring

REFERENCES

- <https://www.youtube.com/watch?v=Dj-X1t5pXvQ>
- Jechow, A., Bumberger, J., Palm, B., Remmler, P., Schreck, G., Ogashawara, I., Kiel, C., Kohnert, K., Grossart, H.P., Singer, G.A. and Nejstgaard, J.C., 2024. Characterizing and Implementing the Hamamatsu C12880MA Mini-Spectrometer for Near-Surface Reflectance Measurements of Inland Waters. *Sensors*, 24(19), p.6445.
- Roman, A., Cluj-napoca, I. D. C. B., Ursu, T., & Cluj-napoca, I. D. C. B. (2018). Multispectral satellite imagery and airborne laser scanning techniques for the detection of archaeological vegetation marks LANDSCAPE ARCHAEOLOGY ON THE NORTHERN AT POROLISSUM. December 2016.



THANK YOU